Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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GitHub Link: https://github.com/atishmandal/CS3807-Deep-Learning-Lab/tree/main/lab8_and_9

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

$$\texttt{daisy}, \quad \texttt{dandelion}, \quad \texttt{roses}, \quad \texttt{sunflowers}, \quad \texttt{tulips}.$$

The images contain variations in:
- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:
- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	Spatial Resolution	Channels
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3 : \text{ after conv4_x},$$

$$P_4 : \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\text{max}} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\text{max}}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

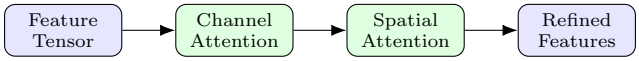
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\, \text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:
- dataset split;

- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	**Attention Position**	**Resolution**	**Channels**
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:
- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:
- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

Is attention at more locations necessarily more effective?

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

Channel → Spatial

against

Spatial → Channel

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time (ms/img)
3×3	0.9055	0.9077	7.97
5×5	**0.9236**	**0.9250**	7.92
7×7	0.8873	0.8863	7.96

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

two additional attention mechanisms

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	**Accuracy**	**Precision**	**Recall**	**Macro F1**	**Parameters**	**Inf. (ms/img)**
No Attention	0.9091	0.9316	0.9056	0.9140	23,597,957	8.02
Channel / SE	0.9200	0.9344	0.9164	0.9223	23,730,117	7.72
Spatial Attention	0.9127	0.9126	0.9157	0.9136	23,598,056	7.65
ECA	0.8982	0.9033	0.8988	0.8983	23,597,962	7.84
CBAM	0.9091	0.9135	0.9139	0.9107	23,730,216	8.25
Self-Attention	0.9418	0.9441	0.9470	0.9444	24,908,678	7.72
Coordinate Attention	0.8945	0.8974	0.8958	0.8931	23,698,469	8.07

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.
The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inf. (ms/img)
Baseline	–	0.9091	0.9316	0.9056	0.9140	23,597,957	8.02
Channel Attention	P_3	0.9200	0.9344	0.9164	0.9223	23,730,117	7.72
Spatial Attention	P_3	0.9127	0.9126	0.9157	0.9136	23,598,056	7.65
Channel+Spatial	P_2	0.9127	0.9225	0.9128	0.9152	23,631,368	7.55
Channel+Spatial	P_3	0.8873	0.8916	0.8925	0.8863	23,730,216	7.96
Channel+Spatial	P_4	0.9018	0.9026	0.9081	0.9016	24,124,520	7.75
Self-Attention	P_3	0.9418	0.9441	0.9470	0.9444	24,908,678	7.72
Additional-1 (ECA)	P_3	0.8982	0.9033	0.8988	0.8983	23,597,962	7.84
Additional-2 (CBAM)	P_3	0.9091	0.9135	0.9139	0.9107	23,730,216	8.25

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.}$$

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score.}$$

This comparison is useful when the class-wise performance is not uniform.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy.}$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:
1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input	Baseline Grad-CAM	Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.

For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:
- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:
- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

30. Qualitative Comparison Across Attention Mechanisms

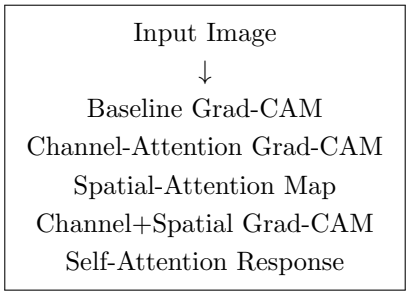
Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:



Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:
- identical training images;

- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

$$\boxed{\text{SE,} \quad \text{ECA,} \quad \text{CBAM,} \quad \text{Coordinate Attention,} \quad \text{Self-Attention}}$$

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. What is the purpose of residual connections in ResNet50?
2. What does conv2_x represent compared with conv5_x?
3. Why does the spatial resolution decrease with network depth?
4. Why does the number of feature channels increase with network depth?
5. What is meant by low-level, intermediate, and high-level features?
6. What is the purpose of an attention mechanism?
7. What does channel attention attempt to learn?
8. What does spatial attention attempt to learn?
9. How is channel attention different from spatial attention?
10. Why can channel and spatial attention be used sequentially?
11. Why might channel attention behave differently at conv3_x and conv5_x?
12. Why can spatial attention be useful at an intermediate ResNet stage?
13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
14. What happens to the size of the self-attention matrix when the spatial resolution increases?
15. How does attention performance vary when the insertion position is changed?
16. How does attention at P_2 differ visually from attention at P_4?
17. Does applying attention at multiple stages always improve classification performance?
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
19. How does the spatial-attention kernel size influence the attention map?
20. How does SE attention differ from ECA attention?
21. How does CBAM combine channel and spatial attention?
22. How does self-attention differ from channel attention?
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
24. What is the difference between an internal attention map and Grad-CAM?
25. Why is Grad-CAM called a post-hoc explanation technique?
26. Does the attention model suppress irrelevant background regions?
27. Are the attention maps of correctly classified and incorrectly classified images different?
28. Which flower classes benefit most from attention?
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?
30. Why is macro F1 useful in addition to accuracy?
31. What is meant by the accuracy–complexity trade-off?
32. What is meant by the F1–inference-time trade-off?
33. Why must every ablation study use the same dataset split?
34. Why must the same training protocol be maintained when comparing different attention mechanisms?
35. What conclusions can be drawn from the Grad-CAM comparison?

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.

13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity}.\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

A. Experimental Setup Actually Used

All results below were produced in a single Colab (T4 GPU) notebook run in a reduced-compute configuration:

- **Data:** `tf_flowers`, 70:15:15 split with seed 42; **50% of each split** was used, giving 1284 training, 275 validation and 275 test images. The same samples were used in every experiment.
- **Training:** Phase 1 – 5 epochs, backbone frozen, learning rate 10^{-3}; Phase 2 – 5 epochs, conv5_x unfrozen, learning rate 10^{-4}. Batch size 32, dropout 0.3, categorical cross-entropy, reduction ratio $r = 16$, 7×7 spatial kernel unless stated.
- **Single run per model** (one seed). One test image equals $1/275 \approx 0.36$ accuracy points, so differences below about 1 point (3–4 images) should not be treated as conclusive.

Feature dimensions immediately before and after every attention block were identical (attention preserves shape):

Position	Before attention	After attention
P_1	$56 \times 56 \times 256$	$56 \times 56 \times 256$
P_2	$28 \times 28 \times 512$	$28 \times 28 \times 512$
P_3	$14 \times 14 \times 1024$	$14 \times 14 \times 1024$
P_4	$7 \times 7 \times 2048$	$7 \times 7 \times 2048$

B. Ablation Results

B.1 Attention-Type Ablation (all at P_3)

Model	Acc.	Prec.	Rec.	F1	Attn. params
A0 Baseline	0.9091	0.9316	0.9056	0.9140	0
A1 Channel	0.9200	0.9344	0.9164	0.9223	132,160
A2 Spatial	0.9127	0.9126	0.9157	0.9136	99
A3 Channel+Spatial	0.8873	0.8916	0.8925	0.8863	132,259
A4 Self-Attention	**0.9418**	0.9441	0.9470	**0.9444**	1,310,721
A5 Combined (C+S+Self)	0.9309	0.9350	0.9348	0.9347	1,442,980

Inference: Self-attention (A4) is the best single module, and the combined module (A5) is second. Channel+Spatial at P_3 fell below the baseline, so a more elaborate module was not automatically better.

B.2 Attention-Position Ablation (Channel→Spatial)

Exp.	Position	Acc.	Macro F1	Attn. params	Inf. (ms/img)
B1	P_1 (56×56)	0.8764	0.8748	8,563	8.60
B2	P_2 (28×28)	**0.9127**	**0.9152**	33,411	7.55
B3	P_3 (14×14)	0.8873	0.8863	132,259	7.96
B4	P_4 (7×7)	0.9018	0.9016	526,563	7.75

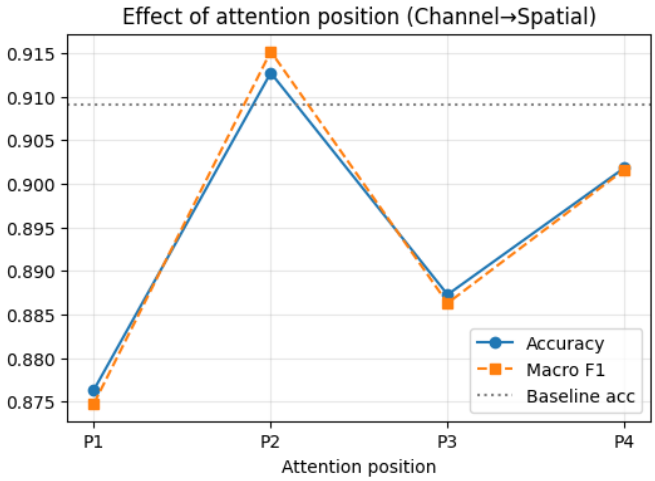


Figure 1: Test accuracy versus the position of the Channel→Spatial module.

Inference: Accuracy does not change in a consistent manner with depth: P_2 is best with 0.913, the highest-resolution P_1 is worst (0.876), and P_3 dips to 0.887. Position therefore mattered.

B.3 Multiple Attention Positions (Channel→Spatial)

Setting	Acc.	Macro F1	Total params	Inf. (ms/img)
P_3 only	0.8873	0.8863	23,730,216	7.96
P_4 only	0.9018	0.9016	24,124,520	7.75
$P_3 + P_4$	**0.9164**	**0.9176**	24,256,779	8.72
$P_2 + P_3 + P_4$	0.8836	0.8853	24,290,190	8.26

Inference: $P_3 + P_4$ improved on either single position, but adding P_2 as well dropped accuracy to 0.884, below the baseline. This shows more attention locations are not always better.

B.4 Attention Order at P_3

Order	Acc.	Macro F1	Inf. (ms/img)
Channel→Spatial	0.8873	0.8863	7.96
Spatial→Channel	0.8982	0.9000	7.83

Inference: Spatial→Channel was about 1.1 points higher (3 images) with identical parameters, a small difference that favours spatial first at P3. The order does affect things.

B.5 Spatial-Kernel Size at P_3

The kernel-size results are given in the table of Section 18: 5×5 was best (0.9236 / 0.9250), 3×3 gave 0.9055 and 7×7 gave 0.8873.

Inference: A medium receptive field (5×5) performed best, and the extremes were worse. The trend seems to suggest that more context from a larger kernel is not automatically useful on a 14×14 map.

C. Graphical Performance Analysis

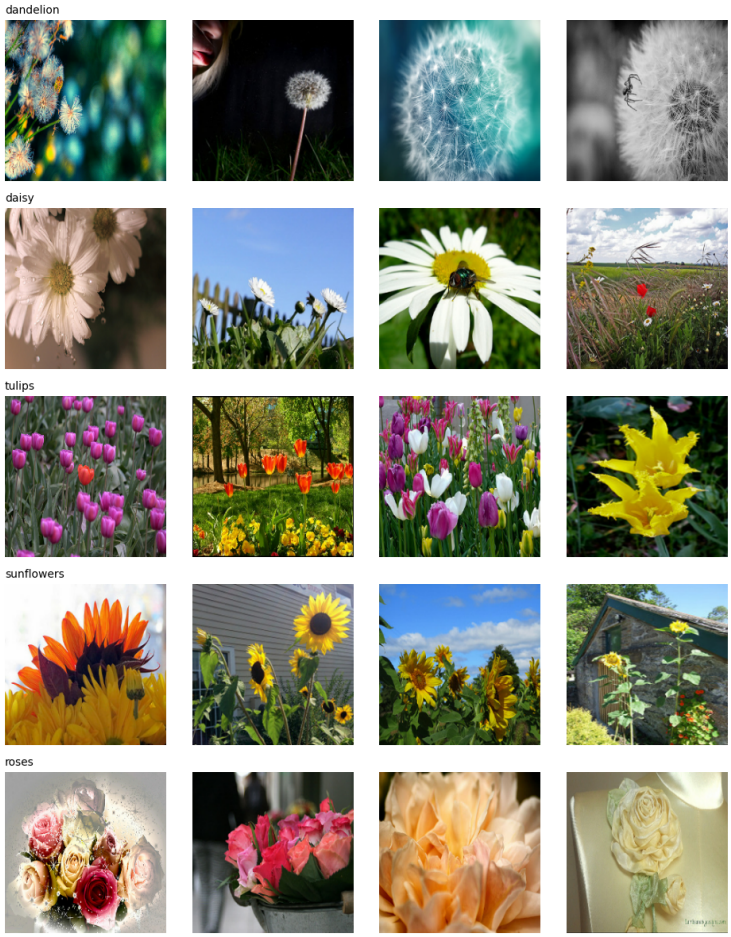


Figure 2: Sample images from all five classes of `tf_flowers`.

Inference: All classes differ in colour and shape (yellow dandelions and sunflowers, white daisies, pink/red roses and tulips), but roses and tulips look slightly similar if at all. Viewpoint and background vary widely, so the dataset is likely to be suitable.

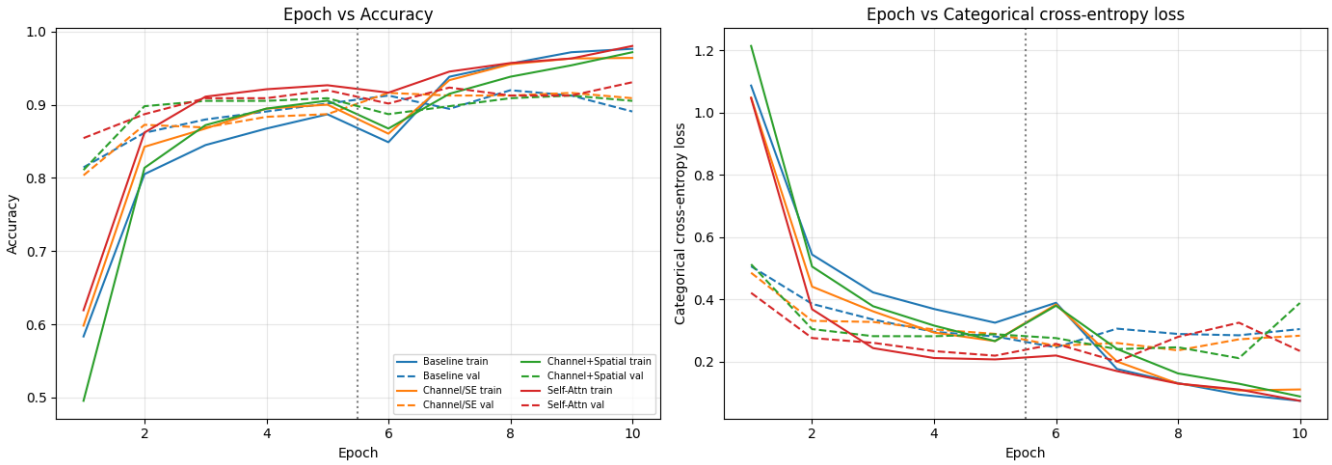


Figure 3: Training (solid) and validation (dashed) accuracy and cross-entropy loss; the dotted line marks the start of fine-tuning.

Inference: Self attention has the highest validation accuracy during phase 1 (reaching 0.92) and the lowest validation loss, so it converged fastest. Training accuracy of all models reaches around 0.96 after fine-tuning while validation stays near 0.89–0.93. Might signal slight overfitting.

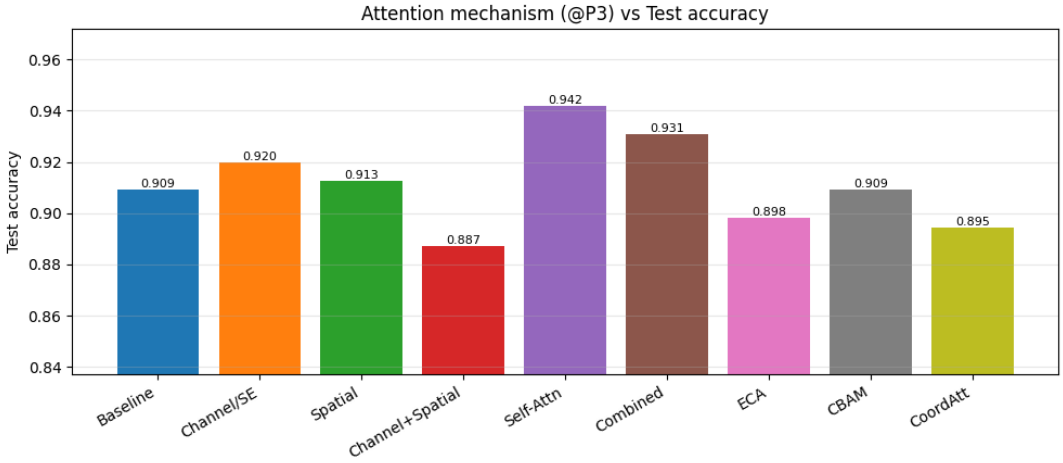


Figure 4: Attention mechanism (at P_3) versus test accuracy.

Inference: Self attention, combined module and channel/SE exceed the baseline.

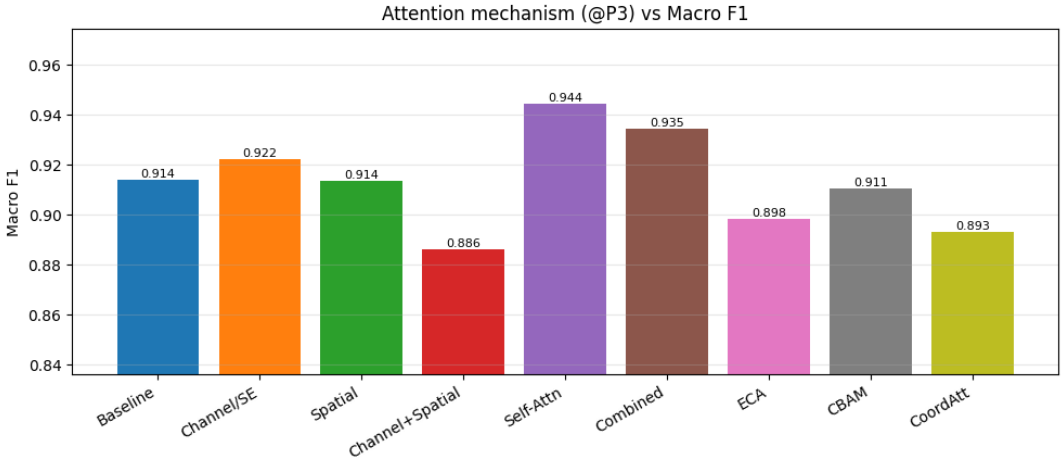


Figure 5: Attention mechanism (at P_3) versus macro F1-score.

Inference: Very similar to accuracy i.e. the changes are not driven by a single weak class.

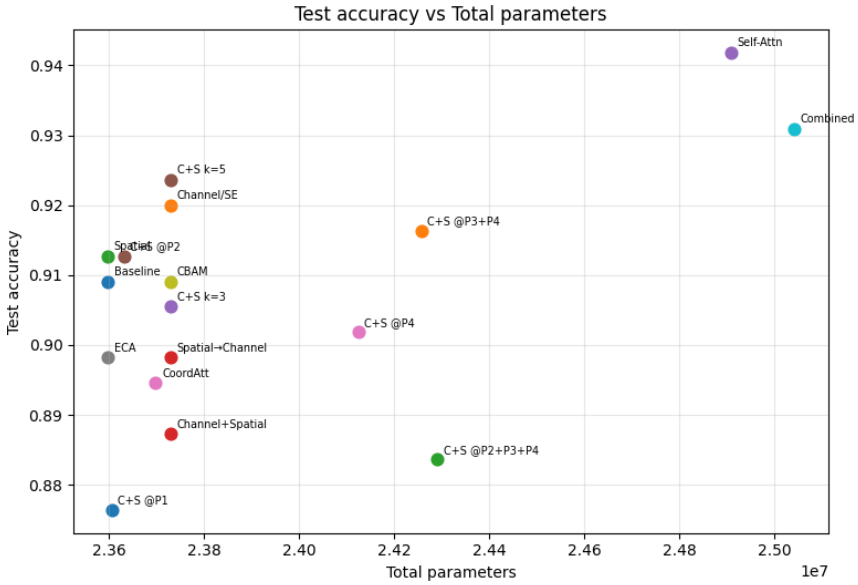


Figure 6: Test accuracy versus total number of parameters for all runs.

Inference: Parameter count is a bad way to guage accuracy - more parameters does not necessarily mean it is better.

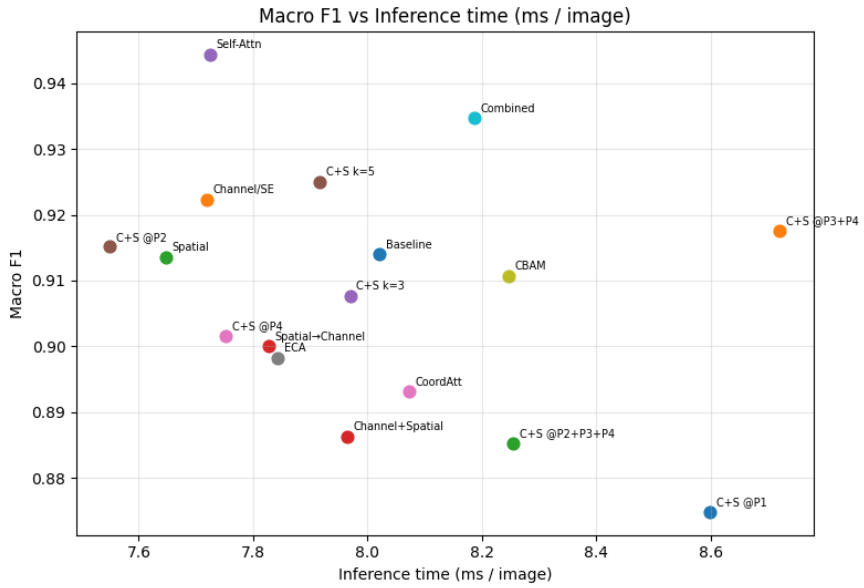


Figure 7: Macro F1 versus inference time per image for all runs.

Inference: Latency differences are small, and the best-F1 models (self-attention, channel/SE) are faster than the baseline. $P_3 + P_4$ and P_1 are slowest without any F1 benefit, so extra computation did not make sense there.

D. Confusion-Matrix Analysis

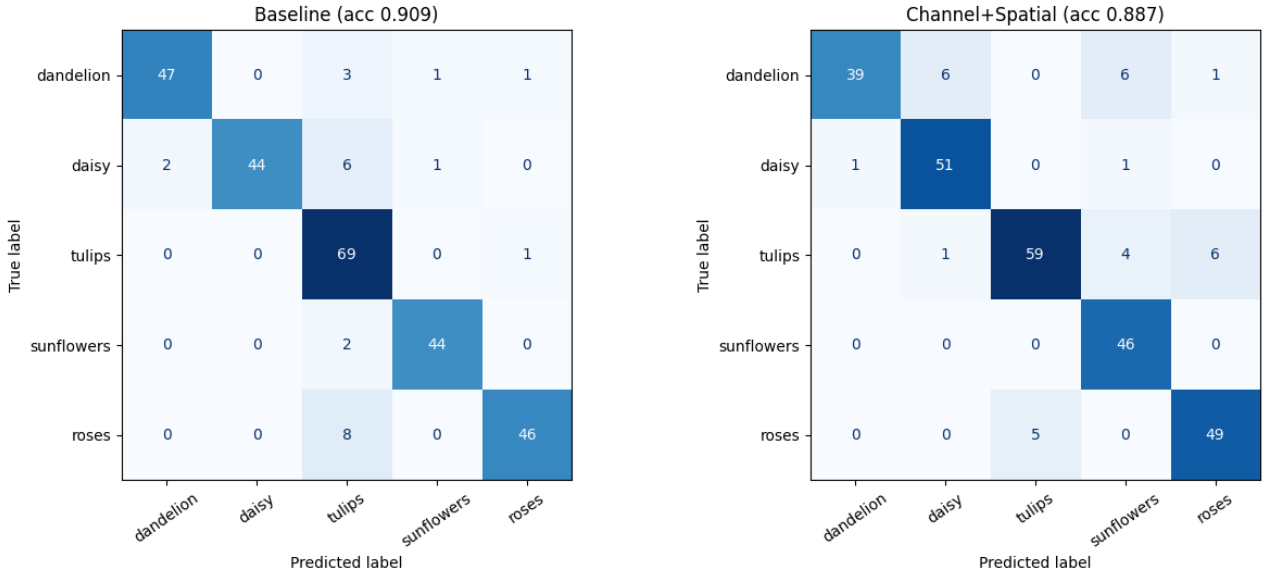


Figure 8: Confusion matrices on the test set: baseline (left) and Channel+Spatial at P_3 (right).

Inference: The baseline's errors concentrate in the tulips column with 8 roses and 6 daisies predicted as tulips; Channel+Spatial reduces these (roses→tulips 8→5, daisy→tulips 6→0, daisy correct 44→51) but introduces new dandelion errors (6 to daisy, 6 to sunflowers).

E. Qualitative Analysis: Internal Attention Maps

Rows in the figures of Sections E and F are the same five test images: three correct (rose, two tulips) and two where the Channel+Spatial model failed (tulips predicted as sunflowers, dandelion predicted as sunflowers).

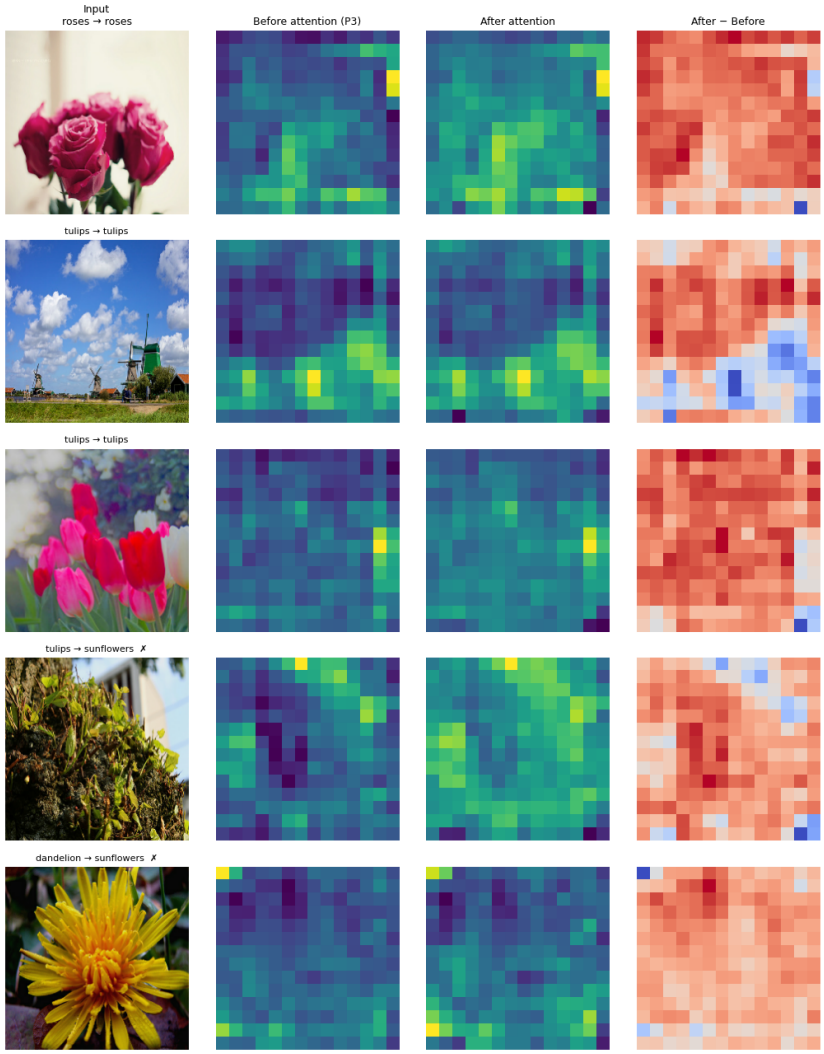


Figure 9: Feature maps at P_3 before attention, after attention, and their difference (Channel+Spatial).

Inference: The hot cells of the feature map are in the same places before and after attention, so the module re weights the features. The difference maps are spread across the whole grid instead of being concentrated on the flower.

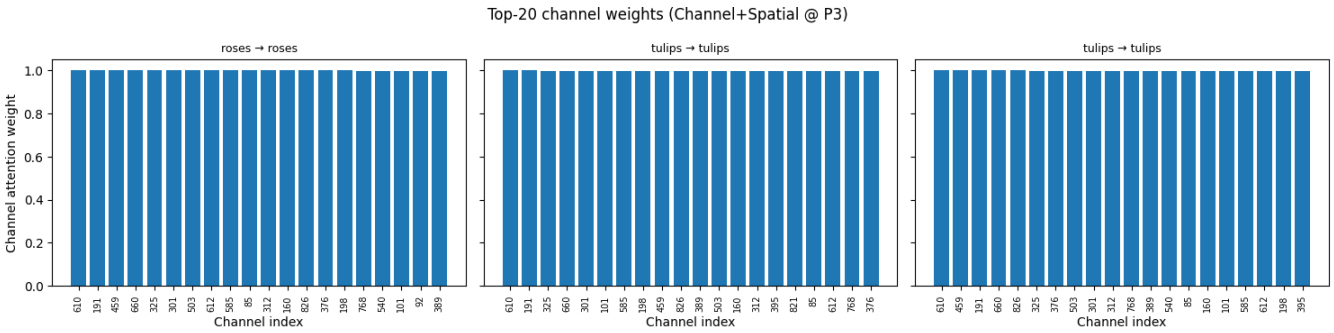


Figure 10: Top-20 channel-attention weights at P_3 for three test images.

Inference: The top 20 weights are all saturated at about 1.0 and the leading channel indices recur across the three images. The weighting is therefore only slightly image-dependent here, with many channels make similar contributions.

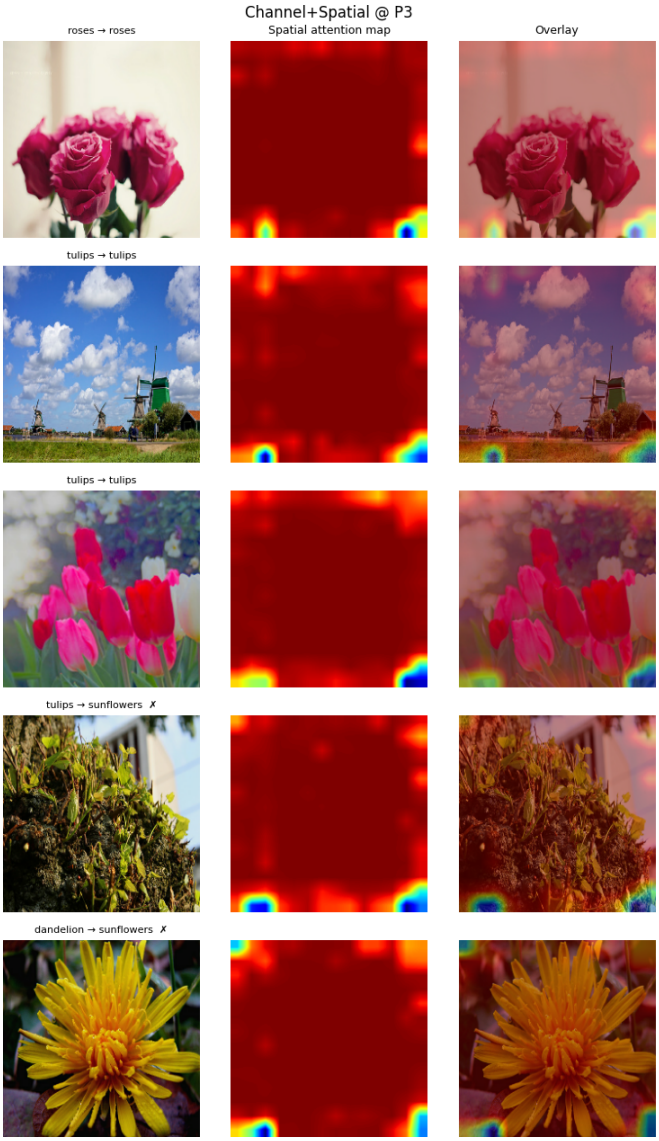


Figure 11: Input, spatial-attention map and overlay for Channel+Spatial at P_3.

Inference: The map is uniformly high over the interior and drops only at border cells. It mainly downweights pixels around the border..

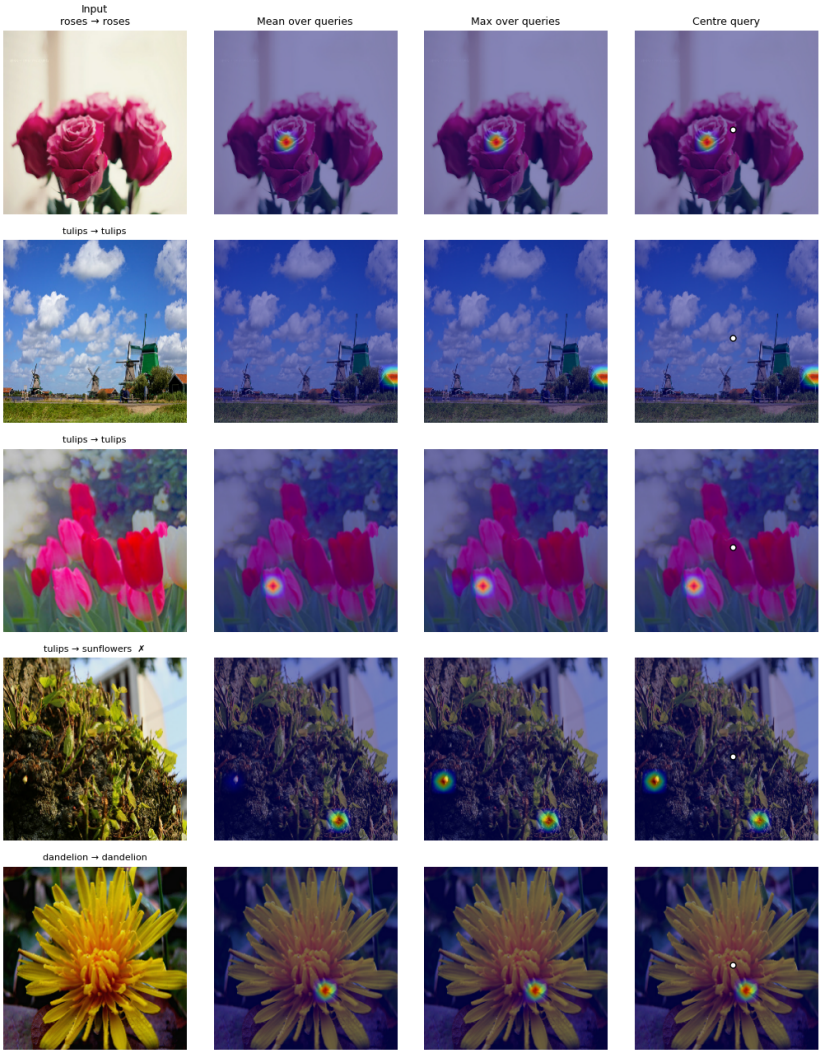


Figure 12: Self-attention response at P_3: mean over queries, maximum over queries and the centre query.

Inference: Each image shows a single compact hotspot, and the mean, maximum and centre query maps are pretty much identical.

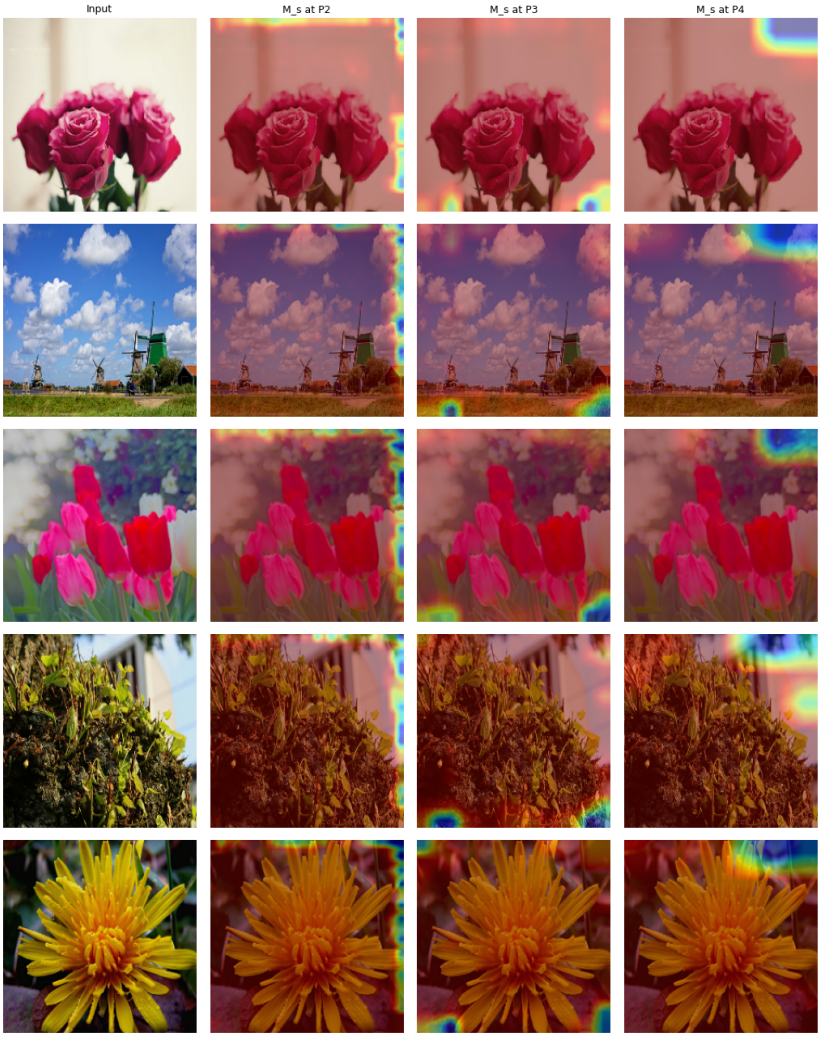


Figure 13: Spatial-attention maps M_s at P_2, P_3 and P_4 for the same images.

Inference: At P_2 the low attention regions are thin strips on the image border. At P_3 they become small blobs in the lower corners. At P_4 a large blob in the top right, so the size of the affected region grows.

F. Qualitative Analysis: Grad-CAM

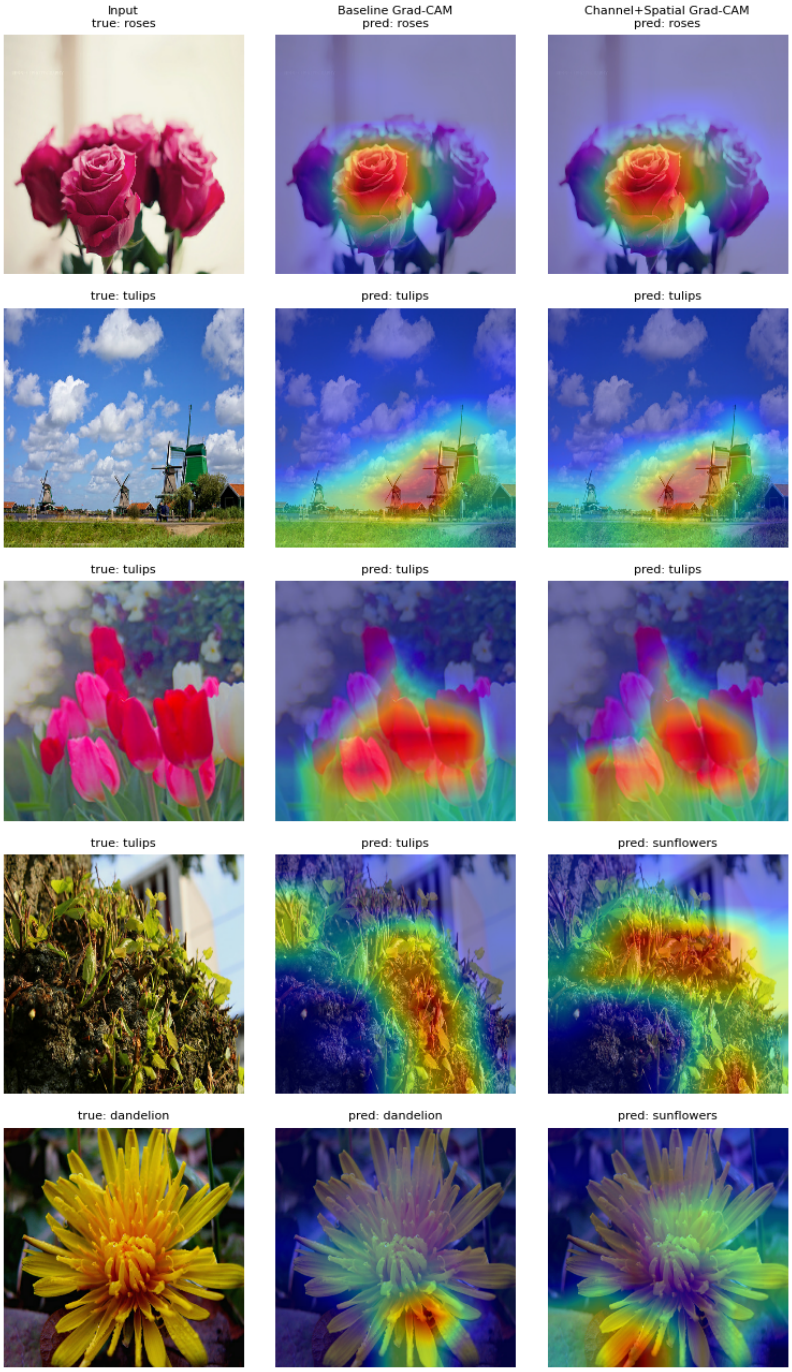


Figure 14: Input, baseline Grad-CAM and Channel+Spatial (P_3) Grad-CAM for the same five images.

Inference: In rose and the two tulip images both models highlight the same flower or flower and grass region and both are correct. For the plant image, the baseline activates the central plant strip while Channel+Spatial shifts to the upper region near the bright background and predicts sunflowers, and for the dandelion it ends up doing opposite it activates the lower petals and predicts sunflowers.

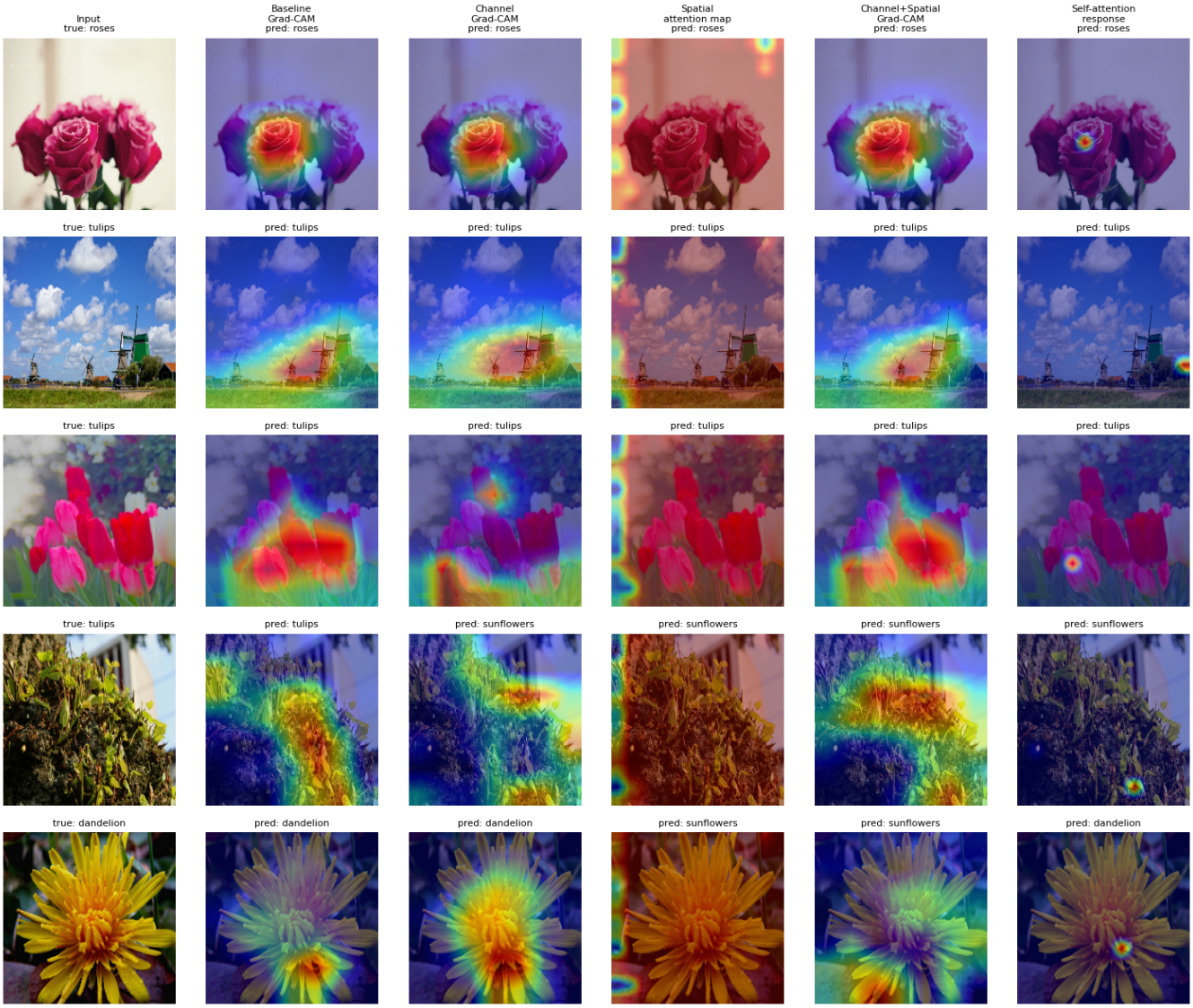


Figure 15: Qualitative comparison: input, baseline Grad-CAM, channel Grad-CAM, spatial attention map, Channel+Spatial Grad-CAM and self-attention response, with predictions.

Inference: All models are correct on the first three images, whereas the plant image is predicted as sunflowers by every attention variant and the dandelion by the spatial and Channel+Spatial models. Only the baseline gets both right and self attention gets the dandelion.

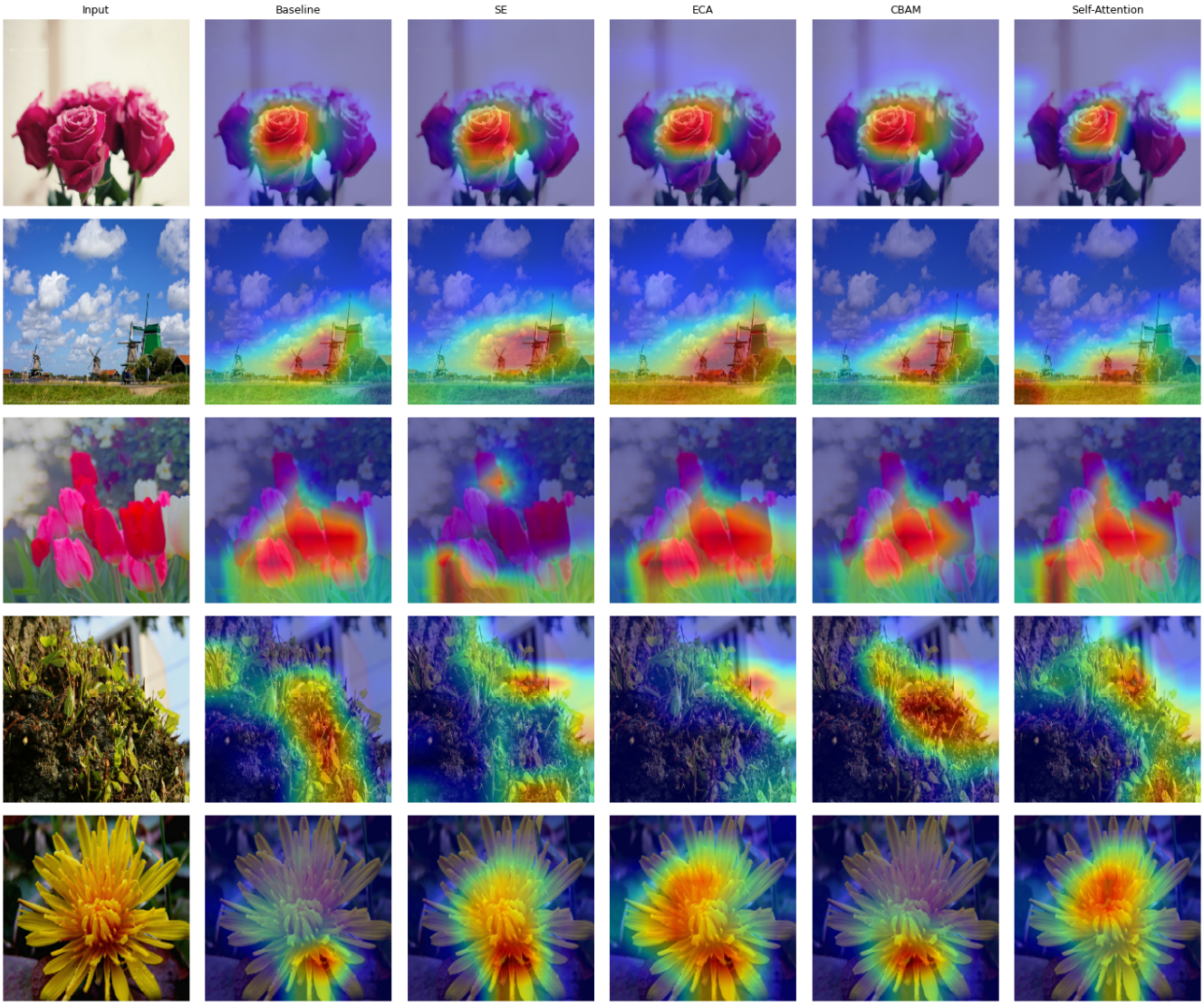


Figure 16: Grad-CAM comparison of baseline, SE, ECA, CBAM and self-attention on the same images.

Inference: On the rose all mechanisms stay on the central bloom, and on the dandelion SE, ECA and self attention cover the whole flower head more broadly than the baseline. Baseline which concentrates on the lower centre; CBAM resembles the baseline.

G. Additional Attention-Mechanism Comparison (Section 19.2)

1. **Accuracy:** 0.887–0.942; Self-attention 0.942, Channel/SE 0.920, Spatial 0.913, CBAM 0.909 (= baseline), ECA 0.898, CoordAtt 0.895.
2. **Macro F1:** 0.886–0.944 with the same ordering as accuracy.
3. **Parameters added:** ECA 5, Spatial 99, CoordAtt 100,512, SE/Channel 132,160, CBAM 132,259, Self-attention 1,310,721.
4. **Inference time:** 7.65–8.25 ms/img; CBAM is the slowest (8.25) and spatial the fastest (7.65), and differences are small.
5. **Convergence:** Only self-attention converged visibly faster (training curves above); the others tracked the baseline.
6. **Feature localization:** SE, ECA and self-attention gave broader coverage on the dandelion, but localization was not consistently tighter than the baseline.
7. **Does complexity always help?** No: Channel+Spatial and CBAM have the same parameter count as SE but are not better; the high-complexity combined module is below self-attention alone.
8. **Lightweight vs heavy:** Channel/SE (0.13M parameters) reached 0.920, near the heavy self-attention (0.942), while ECA with 5 parameters was slightly below the baseline.

H. Answers to Section 37 Questions

1. **Residual connections:** They add the block input to its output, giving gradients a direct path through the network. This allows very deep networks to be trained without the degradation problem.
2. **conv2_x vs conv5_x:** conv2_x is early ($56 \times 56 \times 256$) and holds edges and textures. conv5_x is late ($7 \times 7 \times 2048$) and holds high level semantic information.
3. **Resolution decreases:** Strided convolutions and pooling downsample the map at each stage. This simplifies it and reduces resolution.
4. **Channels increase:** Later stages have to encode more distinct, abstract patterns. Channels are doubled as resolution is halved to keep computation somehwat same.
5. **Feature levels:** Low level features are edges, colours and simple textures. Intermediate features are petals and shapes, and high-level features are class specific object characteristics.
6. **Purpose of attention:** To give a higher weightage to the more informative features and suppress less useful ones.
7. **Channel attention:** It learns which feature channels are important for the input, using pooling. In these maps it kept a near fixed group of channels at about 1.0.
8. **Spatial attention:** It learns which locations are important using channel pooled maps and a convolution. Here it mostly lowered border cells and left the flower high.
9. **Channel vs spatial:** Channel attention answers what (one weight per channel). Spatial attention answers where (one weight per location).
10. **Sequential use:** The two operate on different axes, so they can be used without problems. Channel first selects what matters, then spatial selects where.
11. **Channel attention at conv3_x vs conv5_x:** The channels have different meanings (512 mid-level channels versus 2048 semantic channels). My runs gave different results at P_2 (0.913) and P_4 (0.902).
12. **Spatial attention at intermediate stage:** The map still has enough resolution to localise parts while features are already meaningful. P_2 gave the best Channel+Spatial result (0.913) in these runs.
13. **Self-attention cost:** The attention matrix is $N \times N$ with $N = HW$, so cost grows very fast with the number of positions. At high resolution this is very large.
14. **Matrix size with resolution:** Going from 14×14 (196 tokens, about 38k entries) to 56×56 (3136 tokens, about 9.8M entries) makes it about 256 times larger.
15. **Position effect:** Performance changed with position and was not monotonic: 0.876 (P_1), 0.913 (P_2), 0.887 (P_3), 0.902 (P_4).
16. **P_2 vs P_4 visually:** P_2 shows thin border strips and P_4 larger coarse blobs, consistent with higher versus lower resolution. However neither separated the flower from the background.
17. **Multiple stages:** No. $P_3 + P_4$ (0.916) beat single positions, but $P_2 + P_3 + P_4$ (0.884) fell below the baseline.
18. **C→S vs S→C:** Slightly different: 0.887 versus 0.898 at P_3 with the same parameters. The 3 image gap is small for one seed.
19. **Kernel size:** It sets the neighbourhood used for each map value; larger kernels give smoother, better maps. 5×5 gave the best accuracy (0.924), while 7×7 gave 0.887.
20. **SE vs ECA:** SE uses two fully connected layers with reduction (132k parameters at 1024 channels). ECA uses one small 1D convolution over channels (5 parameters), and was slightly worse here (0.898 vs 0.920).
21. **CBAM:** It applies channel attention (average and max pooling) and then spatial attention to the refined features. It matched baseline accuracy in this run.
22. **Self- vs channel attention:** Self attention relates every spatial position to every other. While channel attention reweights channels from global pooled statistics. Self attention is more accurate but has a lot more parameters.
23. **Largest parameters and best accuracy:** Not necessarily. The combined module (1.44M, 0.931) is below self-attention (1.31M, 0.942), and $P_2 + P_3 + P_4$ (0.69M) is below the baseline.
24. **Internal map vs Grad-CAM:** The internal map is computed in the forward pass as learned weights. Grad-CAM is computed from gradients for a specific class after prediction.
25. **Post-hoc:** It is generated after the model has been trained and has made its prediction, without changing the model. It only explains.
26. **Background suppression:** Spatial attention mainly lowered image borders, and Grad-CAM of the attention model did not always remove the background.
27. **Correct vs incorrect images:** Yes, wrong predictions shifted the Grad-CAM to leaves or bright background, whereas correct ones sat on the flower region.
28. **Classes that benefit:** For Channel+Spatial at P_3, daisy (+0.012 F1) and tulips (+0.007) improved slightly while dandelion and sunflowers worsened. Class benefit depends on the attention model used.
29. **Confusion matrix vs accuracy:** It shows which classes are confused and in which direction (e.g. roses and daisies predicted as tulips). Accuracy only gives a number and does not show class specific misclassifications.
30. **Macro F1:** It averages the F1 of each class equally, so weak classes are not hidden by strong ones. In this case it mirrored accuracy.
31. **Accuracy–complexity trade-off:** Extra parameters should increase accuracy. Channel/SE gives +1.1 points

for 0.13M parameters, while self-attention gives +3.3 for 1.3M. Since compute power is limited, this part has to be carefully decided.

32. **F1–inference-time trade-off:** Higher F1 should justify added latency. In these runs the best F1 models were faster than the baseline, while $P_3 + P_4$ was slower.
33. **Same split:** Otherwise differences could come from the data and not from the model. A fixed split makes the comparision sensible.
34. **Same protocol:** Optimizer, epochs, augmentation and learning rate affect accuracy. Keeping them fixed is obvious for result comparision.
35. **Grad-CAM conclusions:** Models agreed on easy images and diverged on the two hard ones, where the focus moved off the flower.

I. Discussion and Conclusion

Self-attention at P_3 was the best model (accuracy 0.942, macro F1 0.944), followed by the combined module and channel/SE; the other mechanisms were at or below the baseline (0.909). The position study was non monotonic (P_2 best for Channel+Spatial), applying attention at many stages did not help. Evidence showed that spatial attention at P_3 was weakly selective (near uniform maps, nearly input-independent channel weights). .

42. References

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